Practice workbook · Spot constructs, operators, keywords and comments

Reading opened Python code, printed pseudocode or a lettered list of algorithm steps and pointing to an element: a relational or logical operator, the selection keyword, comment text, the line where selection or iteration starts, a line that defines a subprogram, initialises a variable or takes input, including marking such a line with a comment in the editor; also recognising arithmetic, relational and logic operators and why brackets change the order of operations.

23 question parts: 1 worked-example retry and 22 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-13

1. Question 1 · part c.iv · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part c.iv only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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1. Question 1 · part c.iv · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part c.iv only.

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(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

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(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

1. Question 1 · part c.iv · 1 mark · Working and answer

Working and answer

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2. Question 2 · part b.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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2. Question 2 · part b.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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2. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

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3. Question 2 · part c.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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3. Question 2 · part c.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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3. Question 2 · part c.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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3. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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4. Question 2 · part c.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

4. Question 2 · part c.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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4. Question 2 · part c.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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4. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part a · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part a only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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5. Question 1 · part a · 1 mark · Working and answer

Working and answer

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6. Question 1 · part e.i · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.i only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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6. Question 1 · part e.i · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.i only.

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DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

6. Question 1 · part e.i · 1 mark · Working and answer

Working and answer

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7. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.ii only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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7. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.ii only.

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DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

7. Question 1 · part e.ii · 1 mark · Working and answer

Working and answer

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8. Question 1 · part a · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part a only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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8. Question 1 · part a · 1 mark · Working and answer

Working and answer

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9. Question 1 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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9. Question 1 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

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(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

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Working and answer

10. Question 1 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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10. Question 1 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

DO NOT WRITE IN THIS AREA

(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

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Working and answer

11. Question 1 · part d.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iii only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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11. Question 1 · part d.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iii only.

DO NOT WRITE IN THIS AREA

(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

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Working and answer

12. Question 1 · part d.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iv only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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12. Question 1 · part d.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iv only.

DO NOT WRITE IN THIS AREA

(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

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Working and answer

13. Question 4 · part b.i · 1 mark

2022 JUN · Paper 01

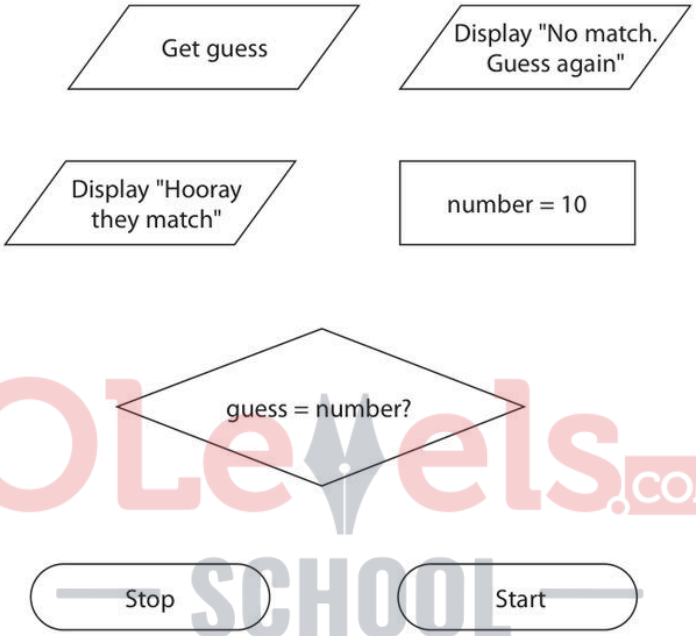
Further practice · Attempt part b.i only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

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P 7 2 5 3 8 A 0 1 0 2 0

13. Question 4 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

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DO NOT WRITE IN THIS AREA

Draw your flowchart here.

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13. Question 4 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

.....

(ii) Give the letter of the step that inputs a value.

(1)

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 2 2 0

13. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

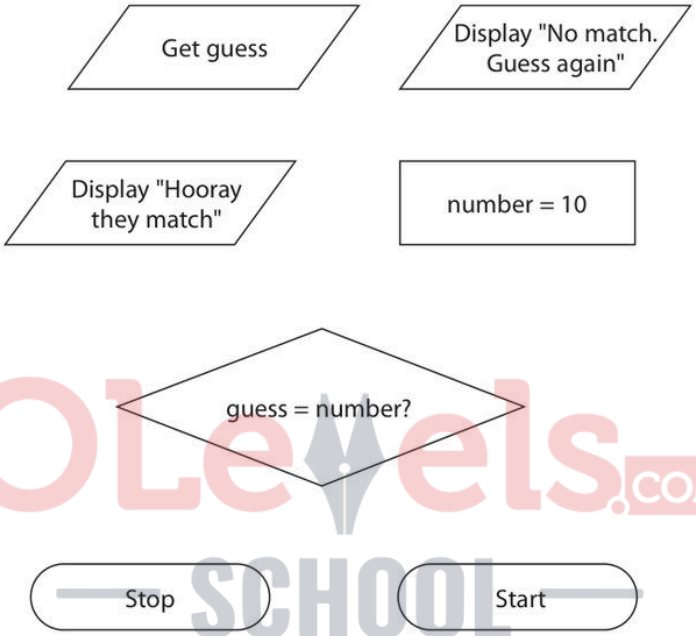
Further practice · Attempt part b.ii only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

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P 7 2 5 3 8 A 0 1 0 2 0

14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

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Draw your flowchart here.

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14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

.....

(ii) Give the letter of the step that inputs a value.

(1)

.....

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14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

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QUESTION 4 (c) BEGINS ON THE NEXT PAGE.

14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

(c) **Figure 2** shows the pseudocode for an early version of an algorithm that Reba has written for another game.

The algorithm:

- asks the user to input a colour or input –1 to end the game
- awards 1 point for red
- awards 8 points for orange
- generates the score for the game
- displays the results of the game.

```
SET Colour TO ""
SET Score TO 0
SET RedPoints TO 0
SET OrangePoints TO 0
SET NumOranges TO 0

WHILE Colour <> "-1" DO
   RECEIVE Colour FROM (STRING) KEYBOARD
   IF Colour = "red" THEN
      SET RedPoints TO RedPoints + 1
   ELSE
      IF Colour = "orange" THEN
         SET OrangePoints TO OrangePoints + 8
         SET NumOranges TO NumOranges + 1
      END IF
   END IF
END WHILE

SET Score TO RedPoints + OrangePoints

SEND ("Score: "& Score) TO DISPLAY
SEND ("Number of reds: "& RedPoints) TO DISPLAY
SEND ("Number of oranges: "& OrangePoints) TO DISPLAY
```

Figure 2

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14

P 7 2 5 3 8 A 0 1 4 2 0

14. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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15. Question 2 · part a.i · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.i only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
    IF pNumber = 1 THEN
        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

DO NOT WRITE IN THIS AREA
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15. Question 2 · part a.i · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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15. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

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16. Question 2 · part a.ii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.ii only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

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FUNCTION checkPrime(pNumber)
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        check = True
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                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

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P 7 2 4 0 6 A 0 4 1 6

16. Question 2 · part a.ii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.ii only.

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DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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16. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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17. Question 2 · part a.iii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.iii only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
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    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

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17. Question 2 · part a.iii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.iii only.

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DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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17. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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18. Question 1 · part b.i · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.i only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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18. Question 1 · part b.i · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.i only.

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DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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18. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

19. Question 1 · part b.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.ii only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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OLevels.com SCHOOL

19. Question 1 · part b.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

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(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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19. Question 1 · part b.ii · 1 mark · Working and answer

Working and answer

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20. Question 1 · part c.i · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.i only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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20. Question 1 · part c.i · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.i only.

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(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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20. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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21. Question 1 · part c.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.ii only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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21. Question 1 · part c.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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21. Question 1 · part c.ii · 1 mark · Working and answer

Working and answer

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22. Question 1 · part c.iii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.iii only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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22. Question 1 · part c.iii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.iii only.

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(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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22. Question 1 · part c.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

23. Question 1 · part d.ii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.ii only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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23. Question 1 · part d.ii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.ii only.

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DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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23. Question 1 · part d.ii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.ii only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

.....

(ii) Identify a logical operator used in this program.

(1)

.....

(iii) Identify the name of a variable that holds a real number.

(1)

.....

(iv) Identify the name of a parameter.

(1)

.....

(Total for Question 1 = 14 marks)

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DO NOT WRITE IN

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Working and answer